

Hip Arthroscopy

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The technique of hip arthroscopy has undergone a period of rapid evolution and growth. In its infancy, hip arthroscopy was mostly for diagnostic and limited therapeutic utility. It was “a technique looking for indications.” Now, improved understanding of intra- and extra-articular hip disease has led to increased investigation of both non- and prearthritic hip disorders. The unique technical challenges to arthroscopic access imposed by the hip joint, different from that of the knee and shoulder, include its deep location surrounded by multiple musculotendinous and neurovascular structures, “ball and socket” type joint congruity requiring traction, and necessity of variable degrees of capsular incision for visualization and instrumentation. Improved equipment and technique has now permitted safer and easier arthroscopic and endoscopic access to virtually any musculoskeletal structure around the hip. Thus hip arthroscopy is now able to effectively diagnose and treat most hip pathology. This largely accounts for the proliferation in performance of hip arthroscopy in the United States and across the world.¹

INDICATIONS

Indications for hip arthroscopy are based on a thorough history, physical examination, and imaging analysis. The Warwick Agreement defined femoroacetabular impingement (FAI) syndrome as a motion-related clinical disorder based on the triad of symptoms (history), clinical signs (physical examination), and imaging findings (plain radiographs and advanced imaging).² Symptoms typically are motion- or position-related hip and/or groin pain. Pain may also be experienced in the back, buttock, or thigh. Mechanical symptoms such as catching, clicking, locking, stiffness, loss of motion, or giving way may also be perceived. A variety of hip impingement tests may be used to elicit the patient’s typical “chief complaint” pain. The flexion-adduction-internal rotation (FADIR) is the most common test for impingement. Restricted hip flexion and internal and/or external rotation are frequently observed. Orthogonal views of the pelvis, including an anteroposterior (AP) and one of many lateral views (e.g., Dunn, frog-leg, cross-table, false profile), are used to identify cam and/or pincer morphology. Confirmation of all three components of the triad during the medical decision-making component of the treatment algorithm for patients with hip pain, in most cases, warrants a trial of nonsurgical measures (rest, activity modification, education, oral nonnarcotic medications, physical therapy, selective intra-articular injections).³

In the event of failure of nonsurgical treatment, patient dissatisfaction with their current hip condition may indicate hip arthroscopy.

The primary indications for hip arthroscopy include FAI syndrome due to either cam and/or pincer morphology, labral injury, femoral head and/or acetabular articular cartilage injury, intra-articular loose bodies (e.g., synovial chondromatosis, localized focal pigmented villonodular synovitis), septic arthritis, and certain types of “extra-articular” impingement (e.g., anterior inferior iliac spine [AIIS] subspine impingement, iliopsoas impingement). Patient selection is critical in recognition of one or more of these indications that frequently coexist. Patient selection is even more critical in recognition of any contraindications that could portend an increased risk of failure, complications, reoperations, or conversion to total hip arthroscopy.

Contraindications

Contraindications to hip arthroscopy include advanced hip osteoarthritis, more than mild or borderline dysplasia (without concomitant simultaneous or staged periacetabular osteotomy), and patients with asymptomatic cam or pincer morphology. Patients with advanced joint space narrowing along the weight bearing source (<2 mm) are at a significantly higher risk of failure of arthroscopy.^{4,5} Other radiographic signs of advanced arthrosis include subchondral acetabular cyst formation,⁶ osteophytes, subchondral sclerosis, and Tönnis 2 or 3 classification. These patients are better treated with either continued nonsurgical measures or arthroplasty. Patients with clinical and radiographic features of significant acetabular dysplasia include those with a lateral center edge angle (and anterior center edge angle) of less than 18 to 20 degrees, a Tönnis angle of greater than 15 degrees (acetabular index), a femoral head extrusion index greater than 25%, a broken Shenton’s line, femoral head migration, percent medialization of the iliofemoral line,⁷ and an elevated FEAR (Femoro-Epiphyseal Acetabular Roof) Index.⁸ Although technically more challenging, patients with protrusio acetabuli (femoral head medial border is medial to the ilioischial line) pincer morphology are not contraindicated and may be successfully managed arthroscopically.⁹ Given the high prevalence of cam (37%; 55% in athletes and 23% in nonathletes) and pincer (67%) morphology in asymptomatic individuals, surgeons must always remember to “treat the patient, not the x-ray.”¹⁰ There is no role for prophylactic arthroscopic hip preservation surgery in asymptomatic individuals.¹¹

Abstract

The technique of hip arthroscopy has rapidly grown and evolved. Improved instrumentation and technical skills now permit safer and easier arthroscopic and endoscopic access to virtually any musculoskeletal structure around the hip. Thus hip arthroscopy is now able to effectively diagnose and treat most hip pathology. This largely accounts for the proliferation in performance of hip arthroscopy in the United States and across the world. Patient selection is critical in optimizing subjective and objective post-operative outcomes. Indications for hip arthroscopy include FAI (femoroacetabular impingement) syndrome due to either cam and/or pincer morphology, labral injury, certain types of extra-articular impingement, articular cartilage injury, and intra-articular loose bodies. Contraindications include asymptomatic cam and/or pincer morphology, advanced arthrosis, and more than borderline dysplasia (without a concomitant periacetabular osteotomy). Operating room setup and positioning are vital components of a well-done supine (or lateral) hip arthroscopy that minimizes risk of traction- and portal-related complications and optimizes outcomes. Surface landmarks, in conjunction with fluoroscopy, can be used to accurately and precisely place portals without iatrogenic chondrolabral or neurovascular injury in all cases. Central compartment management includes labral treatment (repair, reconstruction, débridement), acetabuloplasty rim resection, subspine decompression, femoral head and acetabular articular cartilage management, and loose body removal. Peripheral compartment management includes primarily cam correction. Capsular management includes capsulotomy (interportal with or without T extension) and capsular repair or plication.

Keywords

hip arthroscopy
FAI syndrome
femoroacetabular impingement
labral tear
cam
pincer
capsule

Learning Curve

A steep learning curve is associated with performance of hip arthroscopy. Surgeons must be aware of this learning curve and do as much as possible to reduce the risk of adverse events during the curve. This includes both observation and hands-on training with an experienced hip arthroscopist(s). Attendance at society, institutional, and/or industry cadaveric laboratory workshops are highly valuable in learning the techniques, equipment, and instrumentation. Patient-reported outcome scores,^{12,13} reoperation rate,¹² arthroplasty conversion rate,¹² operative time,^{13–15} complication rates,^{13,14,16–18} failure rates,¹⁵ and patient satisfaction^{12,13} have all been shown to significantly improve with greater surgeon experience. Additionally, the rate of iatrogenic articular cartilage and labral injury during portal placement (the most common hip arthroscopy complication at the most likely time) has been shown to be reduced with greater surgeon experience.^{13–16}

OPERATING ROOM SET-UP AND POSITIONING

Prior to entering the operating room theatre, the patient's correct site (hip) and side (right vs. left) are marked by the surgeon, and confirmed with the preoperative medical records and written informed consent form. A multimodal pain management protocol is begun in the preoperative holding area in order to reduce the need for postoperative narcotic use. The author's preferred "cocktail" includes oral acetaminophen 500 mg, celecoxib 400 mg, pregabalin 75 mg, famotidine 20 mg, and promethazine 12.5 mg; and transdermal postauricular scopolamine 1 mg patch. Upon entry into the operating room, intravenous antibiotic (cefazolin—first generation cephalosporin [1 g if <75 kg body weight; 2 g if 75 kg or greater] or clindamycin [600 mg] if allergic to penicillin or cephalosporin) is commenced within 30 minutes of anticipated skin incision.

Equipment and Instrumentation

Dedicated hip arthroscopy instrument sets are necessary to effectively perform hip arthroscopy. A variety of different arthroscopy towers exist to hold the camera system, camera light source, arthroscopy image/video data source and printer, arthroscopy fluid pump, arthroscopy suction canister and storage source, radiofrequency system, and shaver/burr power system (Fig. 79.1). The video monitor may be on the tower or on a separate ceiling or wall fixture, to visualize the arthroscopy video on one screen and fluoroscopy images on a separate screen. The arthroscopic fluid pump should be at a low pressure (40 to 50 mm Hg) with high flow in order to reduce the risk of fluid extravasation outside the hip joint, potentially into the pelvis, abdomen, retroperitoneum, or chest.^{19,20} The instruments necessary to perform hip arthroscopy may be placed on the scrub technician's back table and at least one Mayo stand (Figs. 79.1 and 79.2A to C). The instruments used for hip arthroscopy, as opposed to knee or shoulder arthroscopy, must be longer in some circumstances. The deep location, thick muscular envelope, and sometimes adipose layer may make standard joint arthroscopy instruments too short to successfully reach the joint. Additionally, the congruent spherical nature of the joint makes curved and flexible instrumentation necessary.

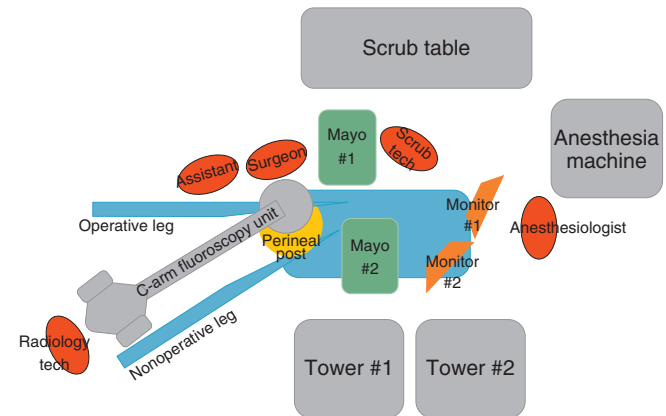


Fig. 79.1 Diagram illustrating room set-up for right-side hip arthroscopy. Note the eccentric positioning of the perineal post to increase the lateral vector upon distraction.

Supine Position

A variety of operating room tables may be used to successfully perform hip arthroscopy. These include fracture tables, joint replacement tables, and hip arthroscopy-specific traction tables. Although the author prefers the supine position, lateral decubitus is also a viable option. The author prefers the Advanced Supine Hip Positioning System with two Universal Hip Distractors and two Active Heel Traction Boots (Smith & Nephew, Andover, MA) and a well-padded perineal post (~25 cm diameter) between the legs. The table has a perineal post holder that is eccentrically located toward the operative side to increase the lateral vector during distraction and reduce pressure on the perineum. Each soft boot is securely placed and Velcro strapped in the Active Heel Traction Boot so that no creases, rivets, grooves, or bulges would cause excessive pressure on each foot or ankle. A self-adherent elastic bandage is placed over the soft boot to increase the surface friction once placed in the Active Heel Traction Boot to reduce the risk of the foot pulling out of the Active Heel Traction Boot.

Examination under anesthesia is performed. This includes measurement of bilateral hip flexion, abduction, and internal and external rotation at 90 degrees hip flexion. Assessment of external rotation recoil, dial testing (maximal limb internal rotation and permissive passive external rotation assessment for endpoint and greater than 45 degrees of external rotation), and traction testing (with or without fluoroscopic assistance) all evaluate capsular (especially the iliofemoral ligament) integrity. Fluoroscopic examination under anesthesia can be helpful to accurately identify trochanteric-pelvic and ischiofemoral impingement. Femoral head translation, in addition to rotation, may also be fluoroscopically observed if the head-neck junction levers on the acetabular fulcrum (microinstability) with or without the presence of a vacuum sign.^{21–23}

Traction

General anesthesia with complete muscle relaxation paralysis reduces the force necessary to obtain sufficient distraction (<50 to 75 pounds) for atraumatic entry into the hip joint.^{24,25} In the lateral position, for every pound increase in distraction

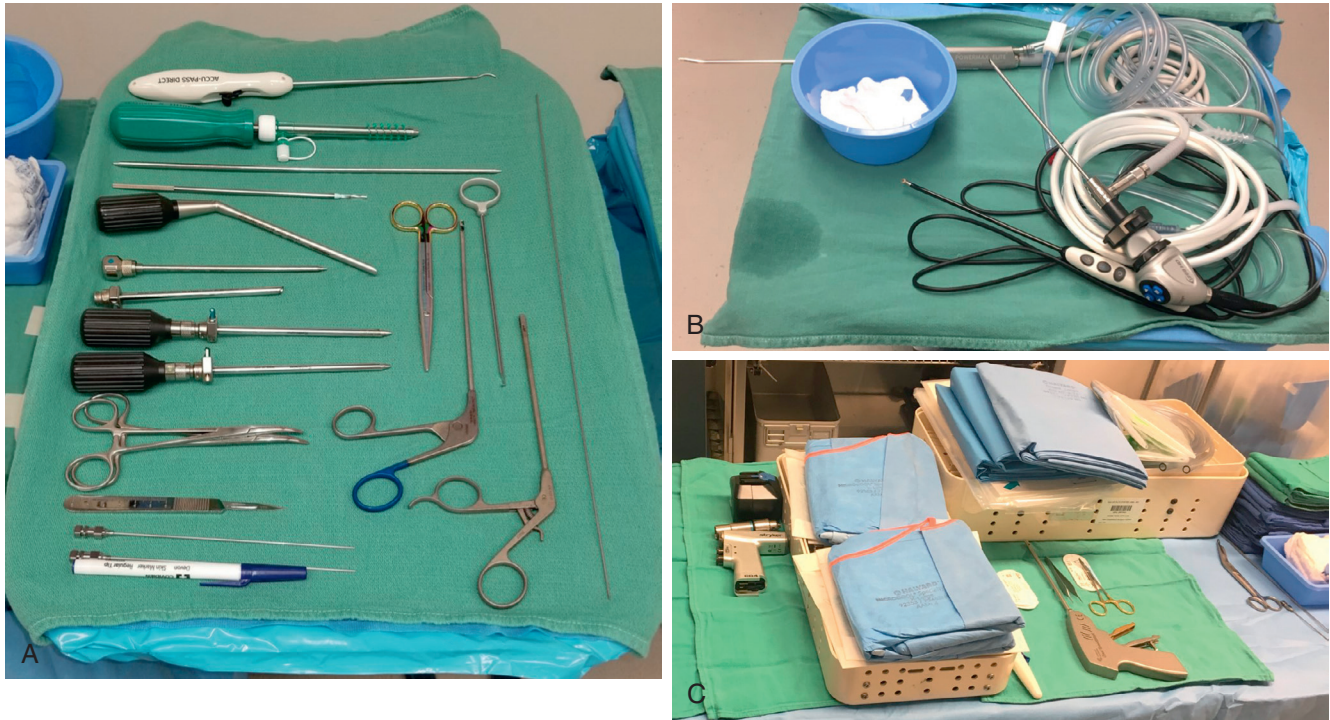


Fig. 79.2 (A) Mayo stand near the surgeon (Mayo #1 in Fig. 79.1) contains (from top to bottom on left side of stand) suture passing device (Accu-Pass Direct, Smith & Nephew, Andover, MA), 8.5 mm clear cannula, switching stick, beaver blade, slotted cannula, 5.0 mm obturator, 5.0 mm cannula, 5.0 mm cannulated obturator in cannula (blue), 4.5 mm cannulated obturator in cannula (green), two hemostats, #11 blade scalpel, two 17-gauge spinal needles, marker; and on right side of stand (from left to right) suture-cutting scissors, looped suture retriever, knot-pusher, arthroscopic suture cutter, and nitinol wire. (B) Mayo stand across table from surgeon (Mayo #2 in Fig. 79.1) contains (clockwise, from top) arthroscopic 4.5 mm curved shaver, arthroscopic fluid tubing, 70-degree arthroscope, radiofrequency wand, and moist lap sponge in fluid bowl. (C) Scrub technician back table contains sterile gowns and glove, two dedicated hip arthroscopy instrument sets, battery power pack for drilling (for suture anchor placement), “shower curtain” sterile adhesive hip arthroscopy drape, nonabsorbable 3-0 suture (for portal incision closure), and sterile towels.

force applied, there is a 4% increase in odds of an adverse nerve event.²⁵ While traction time has not been shown to influence the risk of a postoperative nerve palsy, the duration should be under 2 hours to reduce the risk of a compressive, ischemic “tourniquet-like” effect on the nerves in the perineum.^{25,26} In order to reduce the risk of perineal injury secondary to traction, some authors have performed successful “postless” hip arthroscopy without a perineum or groin complication.^{27,28} The most common complication of hip arthroscopy is iatrogenic articular cartilage and/or labral injury (0.4% to 3.8% and 0.7% to 18%, respectively), occurring primarily during portal placement.^{29,30} Thus patient positioning and traction must optimize the ability to successfully place portals with a reduced, and potentially eliminated, risk of iatrogenic chondrolabral injury.^{31–33}

Once the patient’s feet are securely placed in the traction boots and all bony and soft tissue prominences padded (especially the perineum), distraction commences. A surgical pause “time-out” is performed prior to the initiation of distraction. The nonoperative limb is first placed in approximately 5 degrees of flexion, 45 to 60 degrees of abduction, and neutral rotation. A gentle axial longitudinal distractive pull is applied to provide a counter-traction lateral force to the operative side. The operative limb is placed in approximately 30 degrees abduction and 30

degrees of flexion in neutral rotation and a gentle axial longitudinal distractive pull applied. Next, in a slow, gentle application of zig-zagging hip extension-adduction, extension-adduction, extension-adduction moments, the operative limb is positioned in approximately 5 degrees flexion, 5 degrees adduction, and neutral rotation. The limb is then internally rotated to the degree of femoral anteversion (measured from preoperative imaging, either computed tomography [CT] scan or 3D biplanar fluoroscopy EOS) (if 20 degrees anteversion, then 20 degrees internal rotation). A minimum of 10 mm of distraction is necessary to safely proceed. The C-arm fluoroscopic unit is placed between the legs, roughly parallel with the nonoperative limb, and does not interfere with the surgeon or limb manipulation before or after traction application.

Preparation and Draping

Sterile preparation and draping of the limb with an Isolation Drape with Incise Film and Pouch (3M Health Care, St. Paul, MA) allows for quick and easy application of a surgical drape that protects the sterile field and covers the patient’s body (Fig. 79.3). Using a sterile marker, a straight line is drawn connecting the anterior superior iliac spine and the patella. No portal should be established at or medial to this line. If posterior portals

(posterolateral [PL] or accessory/modified PL) are to be established, the draping must ensure to not block out portal placement posteriorly.

PROCEDURE

Surface Landmarks

Tactile sensation of surface landmarks is used to assist in portal placement during hip arthroscopy. Given the deep constrained nature of the joint, even small errors in portal location can significantly limit access to intra- and extra-articular structures. The anterior superior iliac spine, greater trochanter (tip, anterior



Fig. 79.3 Right hip arthroscopy with sterile preparation and draping. The anterior superior iliac spine is marked with a circle and connected (via *dotted line*) to the patella. No portal or incision should proceed medial to this line. The proximal femur is marked laterally. These surface landmarks are used to estimate the positions of the portals intended to be used.

and posterior margins), and patella are the primary landmarks used. These are reliably and accurately located in most patients' body habitus.³⁴ Traction and adduction may assist in further defining the anterior abductor margin. A 17-gauge spinal needle may be used to estimate the approximate starting point and trajectory of the initial portal placed, the anterolateral (AL) portal. Fluoroscopy is used to ensure correct placement. Over time, both fluoroscopy time and absorbed radiation dose significantly decrease over the first 100 cases of a surgeon's learning curve.³⁵

Arthroscopic Portals

Anterolateral

The AL portal is the initial portal established. The closest structure to the portal is the superior gluteal nerve. However, the nerve is approximately 4.4 cm proximal to the portal.³⁶ A 17-gauge spinal needle is aimed to enter the joint at the 12:30 position on the acetabular clockface (in a right hip). This should roughly parallel the line of the sourcil. The sharp long end of the bevel should be away from the femoral head to avoid iatrogenic articular cartilage damage (Fig. 79.4A). Fluoroscopy is used to assist in AL portal placement. Seldinger technique over a nitinol guidewire is utilized to place a 4.5-mm cannula and introduce the 70-degree arthroscope. Alternatively, some authors recommend initial spinal needle entry into the joint with the bevel facing superiorly to reduce the risk of initial labral injury.^{31,32} An air arthrogram is produced. The capsulolabral interval can sometimes be visualized on the arthrogram (see Fig. 79.4B). Then, the stylet is reinserted and the needle withdrawn to just outside the capsule. Joint venting increases the intra-articular pressure and allows relaxation of the labrum due to the loss of negative intra-articular pressure. The needle may then be reinserted with the bevel facing superiorly to avoid the labrum. As soon as the "pop" is felt, the bevel is rotated 180 degrees so that the bevel is facing inferiorly to reduce risk of iatrogenic femoral head articular cartilage injury. The stylet is removed, a nitinol wire inserted, and the needle removed, followed by dilation, cannulation, and arthroscope

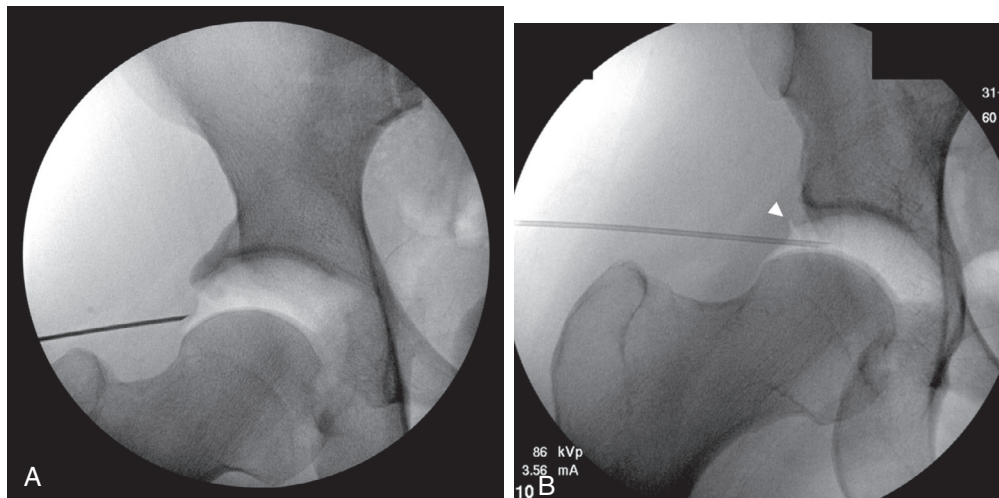


Fig. 79.4 (A) Right hip anteroposterior fluoroscopy image illustrating 17-gauge spinal needle with bevel sharp end away from the femoral head. (B) Right hip anteroposterior fluoroscopy image illustrating an air arthrogram with demarcation of the capsulolabral interval (*white arrowhead*).

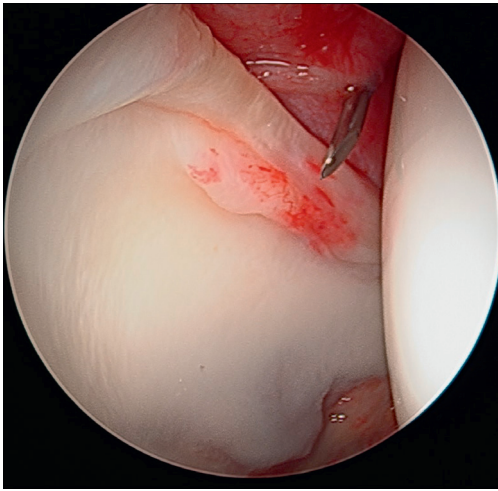


Fig. 79.5 Right hip arthroscopy, 70-degree arthroscope, viewing from anterolateral portal. Placement of spinal needle at approximately 3:00 on clockface for creation of modified mid-anterior portal.

insertion. This method has been shown to reduce the risk of iatrogenic chondrolabral injury to less than 1%.³¹

Modified Mid-Anterior

A modification of the original description of the anterior portal is used. The original anterior portal path is close to the lateral femoral cutaneous nerve (LFCN), within 3 mm.³⁶ Additionally, the portal is within 2 mm of terminal branches of the lateral femoral circumflex artery and within 4 cm of the femoral nerve. The modified mid-anterior portal (MMAP) is lateral and distal to the traditional anterior portal. A superficial skin incision, not penetrating deeply with the knife, is less likely to injure the LFCN. The MMAP is placed in the center of the anterior triangle (Fig. 79.5), entering the joint at approximately the 2:30 to 3:00 position on the clockface, while viewing from the AL portal. Interportal capsulotomy is then created using an arthroscopic beaver blade to connect the AL and MMAP (Fig. 79.6). This greatly enhances visualization and instrumentation in the central compartment.

Distal Anterolateral Accessory Portal

The distal anterolateral accessory (DALA) is placed approximately 4 cm distal to, and in line with, the AL portal. It is most frequently used for percutaneous suture anchor placement. During labral repair, suture anchors are placed in the acetabular rim. Angle of drill and anchor insertion is critically important in ensuring secure anchor placement and avoiding articular cartilage or psoas fossa penetration.^{37–40} Anteriorly, at the 3 o'clock position, the acetabular rim angle is the smallest around the clockface and the amount of bone available is the smallest, making anchor placement challenging (Fig. 79.7).^{41,42} Using the MMAP and a straight drill guide, there is a significantly greater chance of intra-articular penetration.⁴¹ Thus a DALA portal may be used to improve the safety and effectiveness of anchor placement anteriorly to avoid articular cartilage damage. Alternatively, flexible drills and curved suture anchors may be used to circumvent this challenge.⁴³ Use of the DALA portal may be associated with

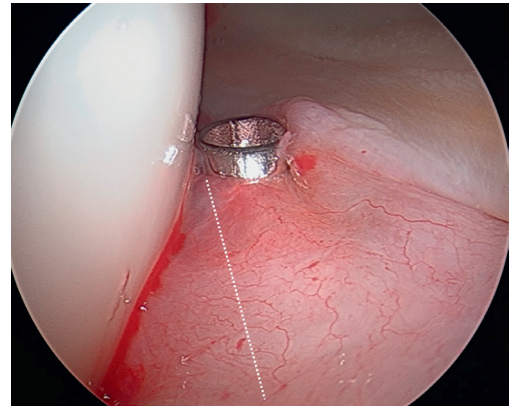


Fig. 79.6 Right hip arthroscopy, 70-degree arthroscope, viewing from modified mid-anterior portal. A 5.0 mm cannula is in the anterolateral portal. Given the known size of the cannula, the interportal capsulotomy can be placed a minimum of 5 to 6 mm from the labrum at the level of the anterolateral portal and move further away from the labrum (while carefully avoiding the femoral head) as it approaches the modified mid-anterior portal. Moving further away from the labrum ensures a sufficient cuff of tissue for interportal capsulotomy repair at the case conclusion.

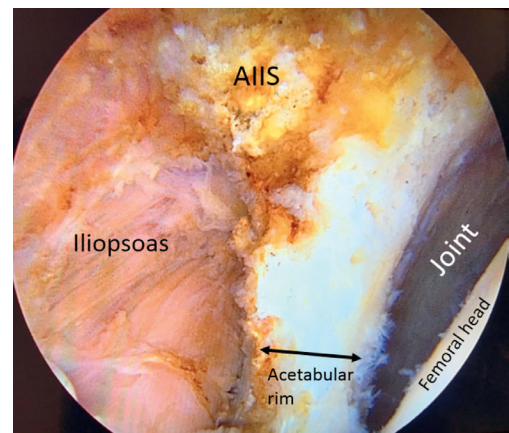


Fig. 79.7 Left hip arthroscopy in cadaveric specimen with labrum removed illustrating the thickness of acetabular rim at the 3 o'clock position. This can be a challenging location for anchor placement if guide starting position, angle of entry, or security of anchor are poor. A/IIS, Anterior inferior iliac spine.

psoas tunnel perforation and persistent pain following labral repair due to the friction associated with the prominent anchor on the adjacent iliopsoas.³⁹

Posterolateral

A PL portal is placed 1 cm posterior and 1 cm proximal to the posterosuperior tip of the greater trochanter, approximately parallel to the same trajectory of the AL portal. This portal may be used for posterior labral repair anchor placement, posterior acetabular rim trimming, or for inflow/outflow fluid management. Although external rotation of the limb moves the sciatic nerve further posterior, it also presents a block (greater trochanter) to proper portal placement. Although internal rotation of the limb moves the sciatic nerve further anterior, the greater trochanter

moves out of the way for portal placement. Neutral limb rotation presents the optimal positioning for PL portal creation.

Technique

Central Compartment

Atraumatic technique during proper portal placement permits an optimally sized interportal capsulotomy to visualize the central compartment, containing the acetabular and femoral head articular cartilage, the fovea and ligamentum teres, the acetabular rim and labrum, and the AIIS. The interportal capsulotomy cuts the fibers of the iliofemoral ligament perpendicularly. The capsular cut should be made with an arthroscopic knife while carefully avoiding accidental articular cartilage injury while inserting the knife. The author does not typically extend the capsulotomy more anterior than 3:00, as this opens the potential for fluid extravasation into the iliopsoas tendon sheath to the pelvis and abdomen. Each case having different requirements, the length of the capsulotomy is variably 2 to 4 cm in length. Posterior to 12:00, the capsule becomes thin and largely deficient blending into the ischiofemoral ligament. From portal placement to interportal capsulotomy creation, the author prefers dry arthroscopy before fluid is introduced and the potential for a “red-out” is reduced. The lateral component of the interportal is made while viewing from the MMAP portal. The medial component of the interportal is made while viewing from the AL portal. After capsulotomy completion, gentle débridement of any frayed ligament fibers using a mechanical shaver or radiofrequency device improves visualization without aggressive capsulectomy. To assist with evaluation of the acetabular rim without capsulectomy, traction sutures placed in each limb of the capsule helps lift the capsule off the rim (“suspension technique”), visualize the capsulolabral interval, and better see and treat acetabular-sided pathology (pincer and subspine impingement).⁴⁴ While an increased length of interportal capsulotomy improves central compartment access and visualization,⁴⁵ it may significantly increase the risk of postoperative rotational and translational hip instability if left unrepaired.^{46–49} The importance of capsular closure is illustrated by improved clinical outcomes with complete closure versus partial or nonclosure.^{50,51} Thus the integrity of the capsule proximal to the interportal capsulotomy must be sufficient to sustain multiple sutures for repair.^{52–54} This emphasizes making the interportal cut far enough away from the labrum (minimum of 8 to 10 mm) to hold the capsular repair: “the key to repairing the capsule is preparing the capsule.”

Adequate exposure of the acetabular rim is necessary to properly address pincer and subspine impingement (Fig. 79.8) and labral tears. Anterior rim visualization (from 1 o'clock anterior to the anterior edge of the transverse acetabular ligament) is improved with use of the AL portal as the viewing portal. Posterior rim visualization (from 1 o'clock posterior to the posterior edge of the transverse acetabular ligament) is improved with use of the MMAP as the viewing portal. Upon identification and characterization of chondrolabral pathology, the decision to perform an acetabuloplasty rim trimming with or without labral takedown/detachment is made. If the labrum is taken down sharply (with a variety of instruments—Bankart elevator, arthroscopic beaver blade, radiofrequency device), rim trimming

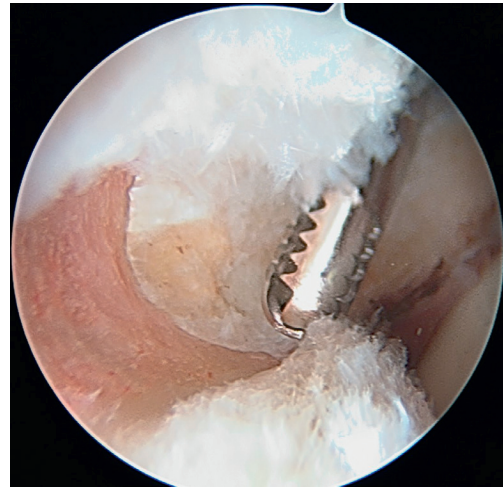


Fig. 79.8 Right hip arthroscopy, anterolateral portal viewing illustrating a completed 2 mm acetabuloplasty rim trimming for focal pincer impingement from 12:00 to 3:00 o'clock and a subspine decompression for a type III anterior inferior iliac spine.

may be visualized better and more precisely treated. This does leave the labrum more unstable than if left attached to the rim. Thus with labral repair/refixation after labral detachment, the potential for overcorrection and labral eversion (and possible suction seal loss) is greater than if left attached to the rim. If the labrum is left attached to the rim, although labral repair/refixation is technically easier and less likely to overly evert the labrum, acetabular rim trimming is more challenging to visualize and treat. Following arthroscopic hip preservation surgery, there is no difference in patient-reported outcome score absolute values or magnitudes of improvement between patients undergoing labral repair/refixation with or without labral detachment.⁵⁵

Preoperative imaging can significantly help in determining the exact magnitude and location of acetabuloplasty rim trimming prior to labral treatment. Standing AP pelvis radiographs best illustrate the lateral center edge angle and standing false profile radiographs best illustrate the anterior center edge angle. There are two different techniques of measurement of these two angles: use of the lateral edge of bone versus the lateral edge of sourcil.⁵⁶ On average, the bone lateral center edge angle is 4 degrees greater than the sourcil lateral center edge angle. On average, the bone anterior center edge angle is 10 degrees greater than the sourcil anterior center edge angle. Thus the surgeon must be cognizant of the exact techniques of measurement pre- and intraoperatively. Once the decision is made to perform an acetabuloplasty, algebraic equations can assist in calculation of the anticipated change in lateral center edge angle (change in angle = $1.8 + [0.64 \times \text{rim reduction in millimeters}]$) based on rim resection at the 12:00 o'clock position.⁵⁷ One millimeter of resection equals 2.4 degrees of lateral center edge angle change and 5 mm of resection equals 5 degrees of change. A standard resection between 12:00 and 3:00 o'clock leads to an anterior center edge angle reduction 1.9 times the magnitude of the lateral center edge angle.⁵⁸ One millimeter of resection equals 2.0 degrees of anterior center edge angle change. The surgeon must be careful to not over-resect. Resection of 4 to 6 mm of rim can increase

contact pressures at the base of the acetabulum by threefold.⁵⁹ Additionally, iatrogenic dysplasia can lead to postoperative instability (dislocation/subluxation).⁶⁰ After the magnitude of bony resection is determined, standard clockface nomenclature can help determine exact location via correlation of preoperative imaging with intraoperative arthroscopic management. The superior margin of the anterior labral sulcus (psoas-u) should be used as 3 o'clock for reference.⁴² Based on this consistent easily identified arthroscopic landmark, the AIIS is typically found at 2:00 to 2:30 and the stellate crease at 12:30. Starting at the capsulolabral interval at the 2:00 to 2:30 position, from deep to superficial, the iliofemoral ligament medial limb (capsule) is first visualized, then iliocapsularis (inferior AIIS facet), and then direct head rectus femoris (superior AIIS facet). Thus if a subspine decompression is to be performed, the capsule necessarily needs to be elevated off the bone in order to visualize the AIIS inferior facet. If performing a subspine decompression, the surgeon must be careful to not inadvertently resect acetabular rim and create iatrogenic anterior undercoverage. Alternatively, a separate superficial to deep trajectory can be used and come on top of the capsule and manage the subspine with a burr from superficial to deep. The surgeon must be careful to not violate the rectus tendon if this latter technique is chosen.

After the rim is managed, the decision to perform labral repair versus débridement is made. The literature is clear that labral repair is better, with significantly greater subjective and objective outcomes.^{61,62} The majority of suture anchors are drilled and placed behind the labrum in the capsulolabral interval. However, alternatively, the thin acetabular rim at the 3 o'clock position may make an internal (in front of the labrum) technique easier (Fig. 79.9). There are two primary techniques for repair available: a looped circumferential suture that goes around the entire substance of the labrum (Fig. 79.10A) or a pierced through-type labral base refixation technique (see Fig. 79.10B) that goes through the mid-substance of the labrum. In a cadaveric model, labral base through-type repairs are significantly better than looped repairs in restoring maximal negative pressure ("suction seal")

and maximal force to distract.^{63,64} Although biomechanical evidence suggests that the suction seal is better retained with a pierced labral base refixation technique, the clinical outcomes between the latter and a looped circumferential repair technique are not significantly different.^{65,66} If labral tissue is deficient (e.g., prior débridement), then a labral reconstruction can significantly improve hip biomechanics (suction seal and resistance to distraction) and patient-reported outcomes.^{63,64,67} Segmental and complete reconstructions similarly improve without significant differences in outcome scores.⁶⁸ There is no difference in outcome improvements based on graft type.^{68,69} The author prefers allograft peroneus longus because of the ease of size matching (approximately 5.5 to 6.5 mm diameter), the ease of manipulation in

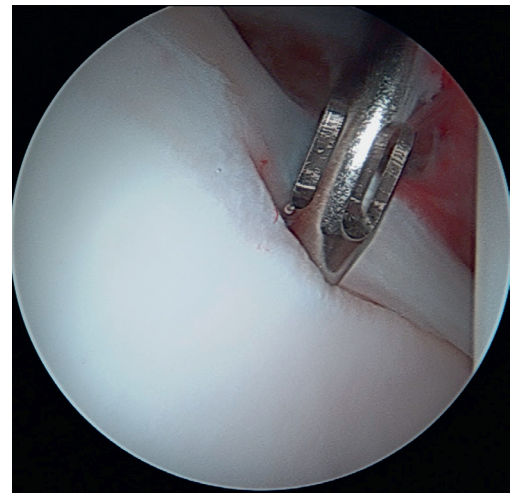


Fig. 79.9 Right hip arthroscopy, anterolateral portal viewing, drill guide for 1.8 mm anchor placement at the 3:30 position. Note guide position in front of the labrum, rather than the more frequently used capsulolabral interval position behind the labrum. The thin acetabular rim between 3:00 and 4:00 o'clock makes the internal position easier for visualization and safe anchor placement.

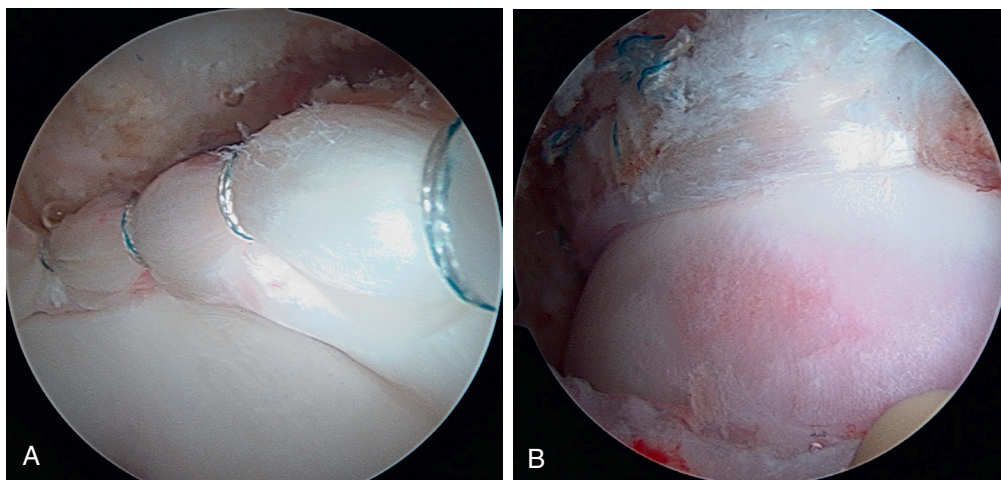


Fig. 79.10 (A) Left hip arthroscopy, anterolateral portal viewing, four suture anchor labral repair with a looped circumferential repair technique. (B) Left hip arthroscopy, modified mid-anterior portal viewing, illustrating the "suction seal" of a pierced through-type labral base refixation technique using four suture anchors.

the joint, the relative lack of swelling in comparison to other graft types, and the minimal preparation time needed.

Peripheral Compartment

Peripheral compartment disorders are managed without traction. Both interportal and T capsulotomies may be created and manipulated to completely visualize and treat all pathologies in the peripheral compartment. These primarily include cam morphology, loose bodies, and synovitis. The author prefers the T capsulotomy as it provides significantly better visualization of the peripheral compartment, with significantly greater cross-sectional area visualization than that of an 8-cm interportal capsulotomy.^{45,70} A precapsular fat pad is frequently present over the fibers of the iliofemoral ligament and must be removed (with a mechanical shaver or radiofrequency device) to expose the interval for the capsulotomy (iliocapsularis [medial] and the gluteus minimus [lateral]).⁷¹ A variety of instruments may be used to create the capsulotomy. The author prefers to view from the MMAP and use a 3.0-mm radiofrequency probe (Dyonics RF Hook 30° Probe; Smith & Nephew, Andover, MA) to create the capsulotomy, bisecting the intermuscular plane down the AL femoral neck to the intertrochanteric line. Traction sutures may be placed in each limb of the “T” to optimize visualization of the head-neck junction, the lateral ascending vessels, and the medial and lateral synovial folds. The author uses a SpeedStitch device (Smith & Nephew, Andover, MA) through the AL portal to place one traction suture in the lateral limb and through the DALA portal to place one traction suture in the medial limb of the “T” capsulotomy. This permits visualization behind the lateral ascending vessels posteriorly and beyond the medial synovial fold inferiorly.

The limb can be manipulated in any position necessary to accomplish a comprehensive cam correction. The author performs the majority of the cam correction in approximately neutral rotation, 5 degrees of flexion, and 10 to 20 degrees of abduction. Intraoperative cam correction should be patient- and hip-specific and based on preoperative imaging. Six intraoperative AP fluoroscopic views (3 views in hip extension [30 degrees external rotation, 30 degrees internal rotation, neutral] and 3 views in 50 degrees flexion [neutral, 40 degrees external rotation, 60 degrees external rotation]) may be used to optimize the correction as these correlate with preoperative 3D CT at the typical cam morphology location from 11:45 to 2:45.⁷² The best intraoperative marker of resection depth is the hard, sclerotic, cortical cam bone. A 5.5 mm round arthroscopic burr is used to perform the cam resection. It is of critical importance to make the correction perfect (Fig. 79.11). A restoration of “ball and socket” mechanics is not achieved if either under- or over-correction occurs. Under-correction (residual cam morphology) is the most common reason for revision hip arthroscopy.⁷³ Over-correction is, in essence, a problem correctable only via arthroplasty. Thus meticulous technique with smooth transition from the proximal head to the head-neck junction, without sharp edges and “shark bites,” is necessary. Both arthroscopic visualization during dynamic impingement maneuvering and fluoroscopic assessment can ensure a proper cam correction. Concurrent use of a 70-degree arthroscope and an AP fluoroscopic image can be challenging at



Fig. 79.11 Right hip arthroscopy, modified mid-anterior portal viewing, illustrating a comprehensive cam correction.

times. If bone resection is being performed and the fluoroscopic image is not changing, the solution is not to keep resecting bone. This is particularly important far laterally and posterolaterally, where the cam morphology is near the vessels and under the labrum. Hip extension and gentle distraction can bring these more proximal lateral cam morphologies into better view. If the osseous morphology extends far beyond the vessels posteriorly or into the vessels, an open surgical approach is indicated. The surgeon must be cognizant of the most proximal extent of the correction and be sure to not over-resect proximally along the entire radial arc of the anterior head-neck junction.⁷⁴ Additionally, distal anterior inferomedial cam is frequently problematic in hypermobile athletes, including dancers, gymnasts, and yogis.⁷⁵ This area must be scrutinized arthroscopically and corrected as this is an area that will mechanically conflict at the AIIS in these patients. The latter illustrates the merit of the T-capsulotomy, as it provides significantly improved visualization and correction in this location. At the conclusion of the cam correction, all bony debris should be evacuated from the joint to minimize the risk for postoperative heterotopic ossification.

Capsular closure commences from distal to proximal along the T capsulotomy. This is a ligament repair. The capsule is the iliofemoral ligament. Thus healing of the iatrogenically violated ligament (the T in parallel and the interportal perpendicular) and subsequent repair is the subject of controversy. The author repairs the capsule in nearly all patients, plicates in those at risk for instability (e.g., hypermobile, Ehlers-Danlos, ballet, borderline dysplasia), and leaves the interportal capsulotomy unrepaired in stiff, tight, male, large cam, Tönnis 1 hips. Both biomechanical and clinical outcomes of hip arthroscopy are significantly better with, rather than without, capsular repair.^{50,76,77} Measurement of generalized joint hypermobility is highly predictive of capsular thickness and a Beighton score of less than four correlates with a capsular thickness of greater than or equal to 10 mm, while a Beighton score of greater than or equal to four correlates with a capsular thickness of less than 10 mm.⁷⁸ Thus in patients with a preoperative Beighton score greater than or equal to four, the

author prefers to plicate, rather than side-to-side repair, the capsule.

Capsular closure is performed using nonabsorbable #2 suture. In the T limb, three or four sutures are typically used, with greater degrees of plication performed in those at risk for instability. In the interportal limb, two to four sutures are typically used, dependent on the size of the interportal capsulotomy made and the degree of inferior capsular shift.⁵⁴ Simple, non-mattress, full-thickness suture technique is used. For the T limb, the arthroscope is in the MMAP and the instrumentation and tying are via the DALA portal. For the interportal limb, the arthroscope is in the AL portal and the instrumentation and tying are via the MMAP. The T limb is closed using a 70-degree up SlingShot (Stryker Sports Medicine, Greenwood Village, CO) to perform a capsular plication. The SlingShot can secure approximately 1 to 2 cm of suture in its penetrating needle and retriever. The surgeon then passes the needle and suture through the lateral limb first (slightly thicker than medial) with an approximately 5- to 10-mm bite of tissue. The suture is deployed deep to the capsule and the needle removed and then repositioned over the medial limb. The needle then pierces and retrieves the suture through the medial limb at approximately 5- to 10-mm bite of tissue. The greater the potential of instability, the greater the plication performed. The author ties each suture, using an initial surgeon's knot, followed by three reversing half-hitches on alternating posts, sequentially, rather than passing all three sutures and then tying. The surgeon then places two to four sutures into the distal limb of the interportal capsulotomy via the DALA portal and each suture is secured with a single hemostat and marked. The arthroscope is then switched to the AL portal. Each distal interportal limb is retrieved through the MMAP and passed through the proximal interportal limb using a SpeedStitch (Smith & Nephew, Andover, MA), proceeding from medial/inferior to lateral/superior. Knots are tied sequentially, the same as the T capsular closure, rather than pass all suture first and then tie. Verification of capsular closure security (Fig. 79.12) is then performed with a dynamic motion examination and all arthroscopic

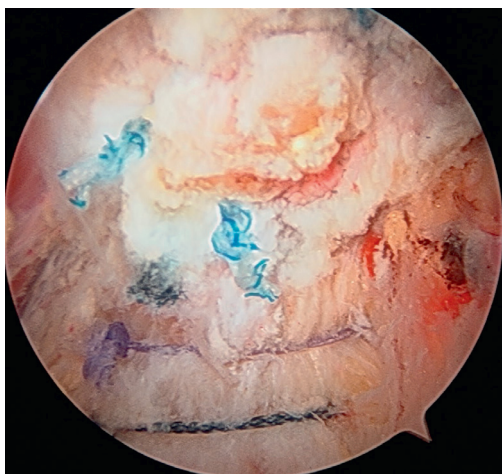


Fig. 79.12 Right hip arthroscopy, anterolateral portal viewing, illustrating a completed capsular closure with three sutures for the T capsulotomy and three sutures for the interportal capsulotomy.

debris evacuated. A postoperative cocktail is placed over the capsule via each portal with a total of 30 mL of 0.5% ropivacaine, 10 mg morphine, and 30 mg ketorolac. Nonabsorbable monofilament suture is used to close the three arthroscopic portals and a sterile dressing applied.

A hinged hip brace (limited from 0 to 90 degrees hip flexion, 0 to 10 degrees abduction) and derotational boots are applied in the operating room. Cryotherapy and compression are recommended for the first 2 weeks. Continuous passive motion machine is recommended for the first two weeks from 0 to 90 degrees hip flexion up to 3 to 4 hours/day while awake. Crutch-assisted foot-flat weight bearing is recommended for 4 weeks postoperatively. Transition to full weight bearing as tolerated commences from week 4 to 6. Patients continue to ambulate with two crutches (without a one crutch transition) until they can successfully walk with a normal gait without a limp. Physical therapy begins preoperatively with patient education and commences postoperatively on day 1 following surgery.

For a complete list of references, go to [ExpertConsult.com](https://www.expertconsult.com).

SELECTED READINGS

Citation:

Frank JM, Harris JD, Erickson BJ, et al. Prevalence of femoroacetabular impingement imaging findings in asymptomatic volunteers: a systematic review. *Arthroscopy*. 2015;31:1199–204.

Level of Evidence:

IV, systematic review of levels I–IV evidence

Summary:

In this systematic review of 26 studies (2114 asymptomatic hips; 57% male; mean age 25 years), the prevalence of cam morphology was 37% (55% in athletes; 23% in nonathletes), pincer morphology was 67%, and labral tear was 68%. This illustrates the high prevalence of abnormal imaging findings in an asymptomatic population, emphasizing the role of treating patients, not x-rays.

Citation:

Griffin DR, Dickenson EJ, O'Donnell J, et al. The Warwick Agreement on femoroacetabular impingement syndrome (FAI syndrome): an international consensus statement. *Br J Sports Med*. 2016;50:1169–76.

Level of Evidence:

Not applicable

Summary:

This international multidisciplinary consensus represented 22 panel members, 5 specialties, and 9 countries. It defined FAI syndrome as a motion-related clinical disorder of the hip with a triad of symptoms, clinical signs, and imaging findings. It encouraged usage of terminology “FAI syndrome,” “cam morphology,” and “pincer morphology” and discouraged usage of “asymptomatic FAI,” “symptomatic FAI,” “FAI morphology,” “deformity,” “lesion,” or “abnormality” referring to cam or pincer morphology.

Citation:

Degen RM, Poultides L, Mayer SW, et al. Safety of hip anchor insertion from the midanterior and distal anterolateral portals

with a straight drill guide: a cadaveric study. *Am J Sports Med.* 2017;45:627–35.

Level of Evidence:

Not applicable

Summary:

This cadaveric laboratory investigation of 16 pelvic specimens examined the safety of suture anchor insertion around the acetabulum (from 9 to 4 o'clock). It showed that anteriorly (2 to 4 o'clock), there is a higher rate of articular surface (4.5%) and psoas tunnel (7.7%) perforation from either the mid-anterior or distal anterolateral accessory portal than previously anticipated.

Citation:

Frank RM, Lee S, Bush-Joseph CA, et al. Improved outcomes after hip arthroscopic surgery in patients undergoing T-capsulotomy with complete repair versus partial repair for femoroacetabular impingement: a comparative matched-pair analysis. *Am J Sports Med.* 2014;42:2634–42.

Level of Evidence:

III, cohort study

Summary:

In this two-year follow-up comparative study of partial versus complete capsular closure by a single surgeon performing

arthroscopic hip preservation, patients that underwent complete repair had significantly better sport-specific outcomes compared with those undergoing partial repair. There was a 13% revision rate in the partial repair group, while no patient needed a revision in the complete repair group.

Citation:

Philippon MJ, Trindade CA, Goldsmith MT, et al. Biomechanical assessment of hip capsular repair and reconstruction procedures using a 6 degrees of freedom robotic system. *Am J Sports Med.* 2017;45:1745–54.

Level of Evidence:

Not applicable

Summary:

This cadaveric laboratory investigation of 10 hip specimens examined range of motion for intact specimens, capsulotomy types (interportal and T), capsular repair, and capsular reconstruction. Both capsular repair and reconstruction may lead to improved patient-reported outcomes by improving rotational hip motion to more near normal in the early postoperative period.

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